

Manideepak Vemuri

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PROFESSIONAL SUMMARY

Experienced Data Engineer with experience building scalable backend APIs, automating ML workflows, and developing production-ready MLOps pipelines. Skilled in Python (FastAPI, Flask), RESTful APIs, PostgreSQL, and cloud-native development using AWS Services. Passionate about integrating machine learning models into real-time systems using tools like MLflow, Scikit-Learn, and Hugging Face. Adept at bridging backend services with intuitive front-end components, driving robust and intelligent product experiences.

EDUCATION

The University of Texas at Dallas <i>Master of Science, Information Technology Management</i> • GPA: 3.361	Jan 2023 - Dec 2024
SRM University, Andhra Pradesh, India <i>Bachelor of Technology, Electronics and Communication Engineering</i> • GPA: 3.4	Jun 2018 - May 2022

TECHNICAL SKILLS

- **Programming Languages:** Python (Pandas, Scikit-learn, NLP), SQL, JavaScript, C#
- **Backend Technologies:** REST APIs (built with Python/AWS Lambda), Flask, Fast API
- **Cloud Platforms:** AWS (S3, Lambda, EC2, Redshift, Glue, Kinesis)
- **Databases:** PostgreSQL, MySQL, AWS Redshift
- **Caching & Messaging:** Apache Kafka, AWS Kinesis
- **DevOps & CI/CD:** Docker, Kubernetes, Terraform, GitHub Actions, Jenkins
- **Monitoring & Debugging:** AWS Cloud Trail, Cloud Watch, Config
- **Version Control:** Git, GitHub, JIRA, Agile/Scrum
- **Computer Science Fundamentals:** Object-Oriented Design, Data Structures, Algorithm Design, Problem Solving
- **Others:** API Development, SQL Query Optimization, Scalability Engineering, Data Preprocessing, ETL Processes

PROFESSIONAL EXPERIENCE

Nagarro <i>Data Engineer</i> • Developed scalable software applications in Python, SQL, and web technologies, contributing to both back-end API development and overall system performance. • Built and optimized SQL-based data pipelines to automate data workflows, significantly improving data processing efficiency and reliability. • Developed cloud-based applications using AWS Services (EC2, S3, Lambda), enhancing system scalability and reliability & supporting back-end infrastructure. • Collaborated with front-end and back-end teams to integrate APIs and improve user interfaces, contributing to the delivery of full-stack web applications. • Participated in the agile development process, contributing to sprint planning, reviews and retrospectives.	Aug 2020 - Dec 2022
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PROJECTS

- Intelligent AI-Powered Recruitment Assistant**
 - Engineered a robust back-end pipeline integrating Gemini and Perplexity APIs for automated and intelligent response generation to job-related communications, enhancing efficiency in recruitment workflows.
 - Developed LLM-integrated (Claude) filters to perform real-time classification, key information extraction, and summarization of resumes and job descriptions, demonstrating advanced ML inference and data preprocessing pipelines.
- Automated Cloud Cost Optimization and Reporting System**
 - Developed a Python-based back-end service to collect and analyze cloud resource usage data from AWS APIs, demonstrating RESTful API interaction and scalable data ingestion.
 - Implemented data preprocessing pipelines to identify cost inefficiencies and anomalies, and generated optimized resource recommendations, showcasing data engineering and analytical skills for automated insights.
 - Designed a responsive front-end dashboard (React) to visualize cost trends and present actionable optimization insights, connecting UI components to back-end APIs and processed data.
 - Automated daily reporting and notification workflows for cost alerts and optimization suggestions, utilizing cloud formation & messaging systems, reinforcing MLOps and Operational efficiency.
- Data Migration Blueprints with AWS**
 - Architected scalable data migration blueprints leveraging AWS services such as S3, Lambda, Glue, and RDS.
 - Developed automated pipelines for secure and seamless data transfers between legacy systems and cloud environment
 - Focused on optimizing performance, scalability, and cost-effectiveness in data migration strategies.
 - Applied Terraform for Infrastructure as Code (IaC) to streamline provisioning and ensure reproducible deployments across environments.
 - Collaborated in an agile team setting, showcasing problem-solving skills and effective communication while aligning deliverables with stakeholder expectations.